Most Common Name?

Most Common Name? 1/2

marquard

zhen

csev

zhen

cwen

marquard

zhen

csev

cwen

zhen

csev

marquard

zhen

Most Common Name? 2/2

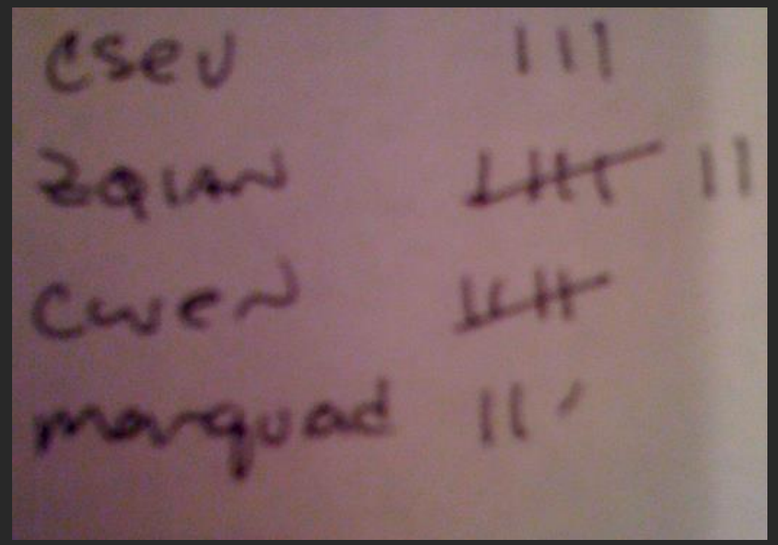
marquard

zhen

csev

zhen

cwen



csev

cwen

zhen

csev

marquard

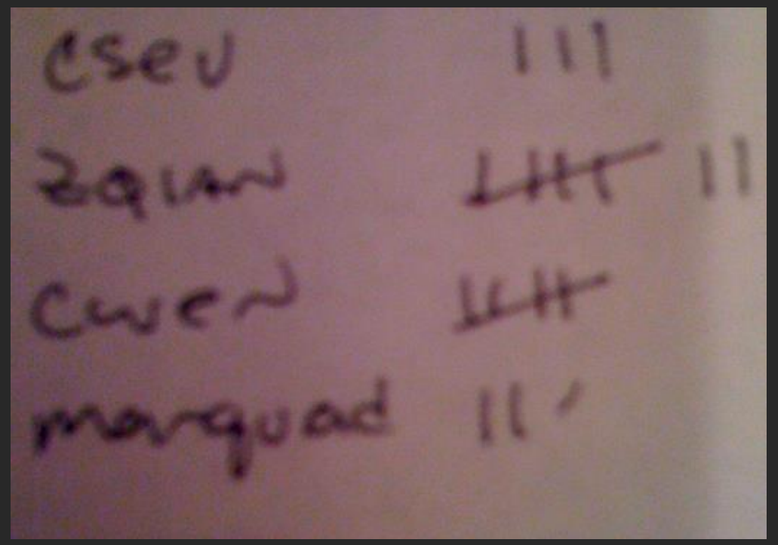
zhen

Many Counters with a Dictionary

One common use of dictionaries is counting how often we “see” something

```
>>> ccc = dict()
>>> ccc['csev'] = 1
>>> ccc['cwen'] = 1
>>> print(ccc)
{'csev': 1, 'cwen': 1}
>>> ccc['cwen'] = ccc['cwen'] + 1
>>> print(ccc)
{'csev': 1, 'cwen': 2}
```

Key Value



Dictionary Tracebacks

- It is an error to reference a key which is not in the dictionary
- We can use the in operator to see if a key is in the dictionary

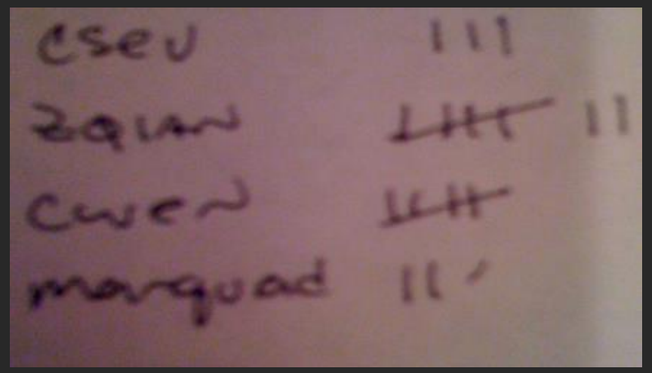
```
>>> ccc = dict()
>>> print(ccc['csev'])
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
KeyError: 'csev'
>>> 'csev' in ccc
False
```

When We See a New Name

When we encounter a new name, we need to add a new entry in the dictionary and if this the second or later time we have seen the name, we simply add one to the count in the dictionary under that name

```
counts = dict()
names = ['csev', 'cwen', 'csev', 'zqian', 'cwen']
for name in names :
    if name not in counts:
        counts[name] = 1
    else :
        counts[name] = counts[name] + 1
print(counts)
```

{'csev': 2, 'cwen': 2 , 'zqian': 1}



The get Method for Dictionaries

The pattern of checking to see if a key is already in a dictionary and assuming a default value if the key is not there is so common that there is a method called get() that does this for us

```
if name in counts:
    x = counts[name]
else :
    x = 0
```

```
x = counts.get(name, 0)
```

Default value if key does not exist (and no Traceback).

{'csev': 2, 'cwen': 2 , 'zqian': 1}

Simplified Counting with get() 1/2

We can use get() and provide a default value of zero when the key is not yet in the dictionary - and then just add one

```
counts = dict()
names = ['csev', 'cwen', 'csev', 'zqian', 'cwen']
for name in names :
    counts[name] = counts.get(name, 0) + 1
print(counts)
```

Default

{'csev': 2, 'cwen': 2 , 'zqian': 1}

Simplified Counting with get() 2/2

```
counts = dict()
names = ['csev', 'cwen', 'csev', 'zqian', 'cwen']
for name in names :
    counts[name] = counts.get(name, 0) + 1
print(counts)
```

Counting is Marvelous
– the Count on Sesame Street